

# Game Name: Ludo King

## 1. Game Overview

**Game:**Ludo King (Online Multiplayer Board Game)

**Core Mechanics:**

Ludo King is an online multiplayer board game played by 2 to 4 players. Each player has four tokens. Players take turns rolling a dice, and the selected token moves forward according to the number rolled. The objective is to move all four tokens from the starting area to the Home zone. The player who successfully brings all four tokens home first wins the game.

## 2. High-Level Architecture

The major systems required to run Ludo King are:

### Matchmaking System

This system connects online players and creates a game room for either 2-player or 4-player matches.

### Turn Management System

This is one of the most important components of the game. It manages player turns, runs a countdown timer for each turn, and automatically passes the turn to the next player when the timer expires.

### Dice & Movement System

This system generates a random number between 1 and 6 and determines whether a player's token is eligible to move according to the game rules.

### Board Logic System

This system manages the entire game board, including safe zones (stars), normal paths, and winning areas. It also handles token captures when one player's token lands on another player's token.

### Game State Management

This system tracks different stages of the match, such as:

- WaitingForPlayers
- GameStarted
- TurnActive
- GameEnded

## 3. Code Structure & Responsibilities

The game follows **Object-Oriented Programming (OOP)** principles and a **Client-Server Architecture**.

## Core Classes

### GameManager (Class)

**Responsibility:**

Acts as the main controller of the match. It manages game initialization, monitors game progress, and determines when the game ends.

### TurnManager (Class)

**Responsibility:**

Controls the turns of Player 1, Player 2, Player 3, and Player 4. It ensures that turns are passed correctly and automatically transfers the turn when the timer runs out.

### Player (Class)

**Responsibility:**

Stores player-related information such as player name, rank, and the collection of four tokens owned by the player.

### Token (Class)

**Responsibility:**

Maintains the current position of each token on the board and handles token movement during gameplay.

## System Communication

The game uses an **Event-Driven Communication Model**.

- When a player clicks the dice button, an event is triggered: `OnDiceRolled(int value)`
- The `TurnManager` receives the dice value and determines whether a token can be moved.
- After the movement is completed, the `Token` class checks whether an opponent's token has been captured (killed).
- The game state is then updated accordingly.

## 4. Design Decisions

### Turn-Based Server System

Ludo King is not a real-time action game like PUBG or Counter-Strike 2. It is a turn-based game, meaning players act one at a time. Therefore, it does not require high-speed servers. The server only needs to synchronize dice results, player actions, and turn information among connected players, resulting in very low bandwidth and data usage.

### Server-Side Verification

The dice value is always generated on the server rather than on the client device. This prevents cheating and ensures fair gameplay. For example, a player cannot manipulate the game to roll a six every turn because all dice outcomes are verified and controlled by the server.